Abhijith Ragav

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Education

Georgia Institute of Technology *Aug 2021 – May 2023*
Master of Science in Computer Science
Courses: *Machine Learning, Artificial Intelligence, Big Data Systems & Analytics, Deep Learning, Ubiquitous Computing*

SRM Institute of Science and Technology *Jul 2016 – Jun 2020*
Bachelor of Technology in Information Technology
Relevant Courses: *Data Structures & Algorithms, Operating Systems, Database Management Systems, Computer Networks, Probability & Statistics, Object Oriented Analysis and Design, Data Mining*

Experience

Amazon, Inc *Seattle, WA*
Applied Scientist — Buyer Risk Prevention *Jan 2024 – Present*
→ Build end-to-end models that protect checkout and prevent fraudulent transactions and bad actors on Amazon.com; blend temporal graph learning over customer–device–payment–address interactions (dynamic embeddings to surface collusive clusters and synthetic identities) with transformer models for tabular/anomaly signals and gradient-boosted trees for calibrated decisioning, plus Bedrock-powered LLM workflows for investigator assist.
→ Delivered business outcomes including 9% reduction in bad debt and 15% fewer orders flagged for manual investigations, improving investigator efficiency and handle time.

Aeva, Inc *Mountain View, CA*
Software Engineering Intern — Perception *Aug 2022 – Dec 2022 | Jul 2023 – Dec 2023*
→ Automated querying trims of data from a log recorded from an ARIES 2 sensor based on a scene description.
→ Implemented self-supervised training to use native ARIES 2 data to classify objects for autonomous driving.
→ Worked with translating synthetic data of specific scenes to a trainable MMLAB format to further improve model performance on pedestrian data.

Amazon *Seattle, WA*
Applied Scientist Intern *May 2022 – Aug 2022*
→ Designed a neural-network-based federated learning framework that predicts buyer fraud across regions, abiding by local data protection regulations.

Carnegie Mellon University *Pittsburgh, PA*
Research Assistant (Synergy Labs) *Jan 2020 – Mar 2021*
→ Built AppService, an infrastructure to host and run IoT apps in discrete LXC containers with dynamic resource allocation.
→ Designed several REST APIs for external systems to deploy, manage and interact with third-party apps on AppService.
→ Interfaced AppService with Building Depot, a Smart-Building OS to interact with 500+ sensors across CMU campus.
→ Worked on a privacy wrapper that restricts third party API calls in an app, and enforces User Access Control.
→ Developed IoT apps that use Machine Learning to make real-time inference on sensor data with minimal overhead.

Solarillion Foundation *Chennai, India*
Undergraduate Research & Teaching Assistant, Server Administrator *Apr 2018 – May 2020*
→ Proposed a scalable deep learning solution for stress detection on edge devices, achieving 95.39% accuracy — absolute increase of 10% over the benchmark and an 18x reduction in inference times on a Raspberry Pi 3.
→ Led a team of 4 and worked with real-world theatre occupancy data to build a two-stage model that predicts the number of weeks a movie is expected to screen based on behavioral population analysis; solution used by a top multiplex in India.

Selected Publications

1. ***REMI: A Novel Causal Schema Memory Architecture for Personalized Lifestyle Recommendation Agents***
   PDF arXiv
   *Workshop paper at **KDD** Workshop on Online and Adaptive Recommender Systems (OARS'25), Toronto, Canada (Aug 6, 2025)*
2. ***Bayesian Active Learning for Wearable and Mobile Health*** Slides/Poster
   NeurIPS Europe Meetup on Bayesian Deep Learning (BDL'20), December 2020 — Poster/Talk
3. ***Bayesian Active Learning for Wearable Stress and Affect Detection***
   *Poster at **NeurIPS** Workshop on Machine Learning for Mobile Health (ML4MH) 2020*
4. ***A Two-Stage Machine Learning Approach to Forecast the Lifetime of Movies in a Multiplex*** Code
   ***FICC** 2020, San Francisco, USA*
5. ***Scalable Deep Learning for Stress and Affect Detection on Resource-Constrained Devices*** Slides
   Code
   ***ICMLA** 2019, Florida, USA*

6. ***Diagnosing Abnormality of Foetus Using Machine Learning Algorithms***
   ICIoT 2019 *— International Conference on Internet of Things, SRM Institute of Science and Technology, Chennai, India (Mar 11–15, 2019)*

Selected Projects

ETHViz: Ethereum Network Topology Visualization & Private-Net GUI Code *2023*
→ Discovered peers from a custom `geth` light node via Web3's `admin.peers()`, geolocated IPs, and rendered interactive maps/graphs (React, Chart.js, D3-geo, React-simple-maps, React-force-graph-3d) backed by a Flask service on AWS EC2; also built a GUI to connect peers/send transactions on private networks.

BERT Siamese Networks for Similar-Intent Question Identification Code Project Blog *Oct 2021*
→ Fine-tuned transformer backbones (BERT, GPT-2, XLNet) on Quora Question Pairs (404k pairs) with transitive augmentation (+25% data); final BERT model reached ~86% test accuracy. Built an unsupervised pipeline (BERT embeddings → PCA to 15 comps, K-means ~100 clusters) to narrow candidates, yielding ~100× faster inference for duplicate detection at scale.

WaDeNet: Wavelet Decomposition based CNN for Speech Processing arXiv *Nov 2020*
→ Proposed an end-to-end speech model that embeds discrete wavelet decomposition within the CNN, enabling the network to learn multi-resolution spectral features without separate feature pipelines; reported an average +6.36% accuracy gain over prior baselines on mobile-health speech tasks with reduced model size/compute.

Transfer Learning for International Crisis Response Leaderboard Code *Jan 2020*
→ Used RoBERTa to transfer knowledge across organizations for crisis-tweet classification; focused on boosting performance for data-sparse orgs; high-ranked on the AMLD 2020 AIcrowd leaderboard.

Pollen Grain Classification Google Scholar Code *May 2020 – Jun 2020*
→ Implemented a two-stage pipeline using U-Net (segmentation) and VGG-16 (classification) to classify microscope pollen-grain images; presented as part of the ICPR 2020 Pollen Grain Classification Challenge/Workshop.

Skills

Languages: Python, C/C++, SQL, Bash, JavaScript
Core ML: transformers for tabular & sequence data, gradient-boosted trees (XGBoost/LightGBM), anomaly detection, self-supervised representation learning, metric learning (Siamese), segmentation/classification (U-Net, VGG), Bayesian active learning, calibration/thresholding
Graph & Network ML: entity/interaction graphs, time-aware representation learning, link prediction, community detection, graph-feature engineering; tools: PyTorch Geometric, DGL, NetworkX
LLMs & GenAI: Amazon Bedrock (agents, knowledge bases, guardrails), prompt design/evaluation, retrieval-augmented workflows, investigator assist
Experimentation & MLOps: offline/online eval, A/B testing, drift monitoring, feature pipelines, model serving & latency/SLA optimization; tools: PyTorch, Hugging Face Transformers, scikit-learn, ONNX, Docker, GitHub CI
Data & Back-end: Pandas, NumPy; FastAPI/Flask; SQL; large-scale feature engineering on logs/graphs
Web3 & Visualization: Web3/geth (ETHViz), React, D3/Chart.js dashboards
Cloud (AWS): Bedrock, SageMaker, EC2, S3, Lambda; basic IAM; cost/performance tuning

Achievements and Extra-curriculars

→ Submission ranked in top 3% in the Spotify Sequential Skip Prediction challenge conducted by Spotify and AICrowd.
→ Best project award at SRM ICIOT 2019.
→ Led the blogging team at SRM Alumni association.